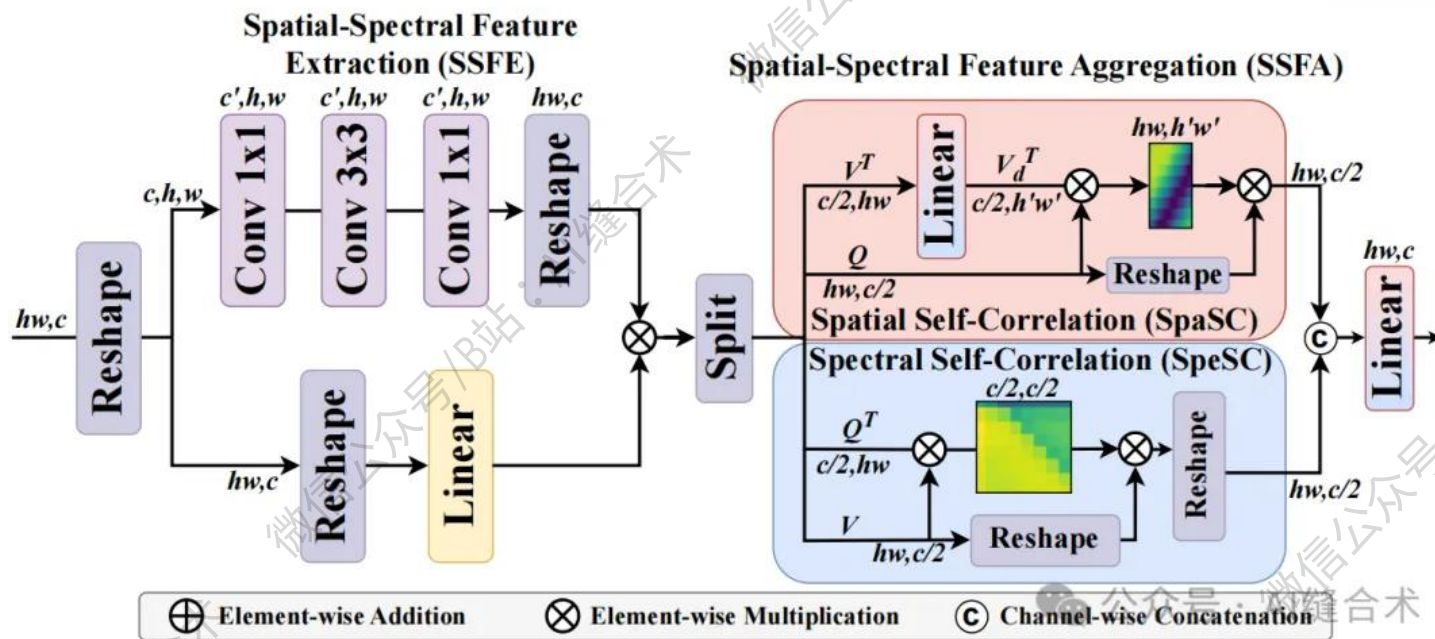


核心速览



论文题目: HSSDCT: Factorized Spatial-Spectral Correlation for Hyperspectral Image Fusion

中文题目: HSSDCT: 高光谱图像融合中的因子化空间-光谱相关性

论文链接:

<https://arxiv.org/pdf/2602.00490>

所属单位: 国立成功大学等

核心速览:

本文提出 HSSDCT 框架,旨在通过融合低分辨率 HSI(LR-HSI)的丰富光谱信息与高分辨率多光谱图像(HR-MSI)的精细空间细节,重建高质量高分辨率 HSI(HR-HSI)。该框架包含两个核心模块:(i) 层级密集残差

Transformer 块 (HDRTB) , 通过渐进式扩大窗口并引入密集残差连接, 实现多尺度特征聚合; (ii) 空间-光谱相关性层 (SSCL) , 显式因子分解空间与光谱依赖关系, 降低自注意力的线性复杂度, 同时缓解光谱冗余。实验表明, HSSDCT 在多个基准数据集上实现了优于现有方法的重建质量, 并显著降低了计算成本。

<https://mp.weixin.qq.com/s/zwl7NXgyARDF6Q5T9UBbfQ>

```
SCC(
  dim=32, window_size=(16, 16), num_heads=4
  (qv): DFE(
    (conv): Sequential(
      (0): Conv2d(32, 6, kernel_size=(1, 1), stride=(1, 1))
      (1): LeakyReLU(negative_slope=0.2, inplace=True)
      (2): Conv2d(6, 6, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
      (3): LeakyReLU(negative_slope=0.2, inplace=True)
      (4): Conv2d(6, 32, kernel_size=(1, 1), stride=(1, 1))
    )
    (linear): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1))
  )
  (proj): Linear(in_features=32, out_features=32, bias=True)
  (value_drop): Dropout(p=0.0, inplace=False)
  (proj_drop): Dropout(p=0.0, inplace=False)
  (spatial_linear): Linear(in_features=1, out_features=1, bias=True)
  (pos): DynamicPosBias(
    (pos_proj): Linear(in_features=2, out_features=2, bias=True)
    (pos1): Sequential(
      (0): LayerNorm((2,), eps=1e-05, elementwise_affine=True)
      (1): ReLU(inplace=True)
      (2): Linear(in_features=2, out_features=2, bias=True)
    )
    (pos2): Sequential(
      (0): LayerNorm((2,), eps=1e-05, elementwise_affine=True)
      (1): ReLU(inplace=True)
      (2): Linear(in_features=2, out_features=2, bias=True)
    )
    (pos3): Sequential(
      (0): LayerNorm((2,), eps=1e-05, elementwise_affine=True)
      (1): ReLU(inplace=True)
      (2): Linear(in_features=2, out_features=4, bias=True)
    )
  )
)
input_tensor_shape : torch.Size([2, 32, 128, 128])
output_tensor_shape : torch.Size([2, 32, 128, 128])
```

哔哩哔哩/微信公众号: AI缝合术, 独家整理!